

DiaTool-DPO:

Enhancing Tool-Augmented LLM Dialogue via Direct Preference Optimization

Automatic paired-trajectory DPO — no human labeling — lifts slot-filling accuracy by +43.5% and tool-rejection by +10.5% on LLaMA-3-8B.

Audience: ML Researchers · NLP Engineers

Task: Tool-Augmented LLMs · RLHF Alignment

Method: 5-State MDP + DPO

SFT Alone Fails to Teach Slot-Filling and Tool Rejection

Core motivation for DiaTool-DPO

01

Failure to Ask Follow-Up Questions

When a user query is missing required arguments, SFT-trained TA-LLMs often skip slot-filling and attempt a tool call with incomplete parameters.

02

Hallucinated Tool Invocations

Without explicit preference signal, the model fabricates argument values rather than asking the user — producing incorrect or dangerous tool calls.

03

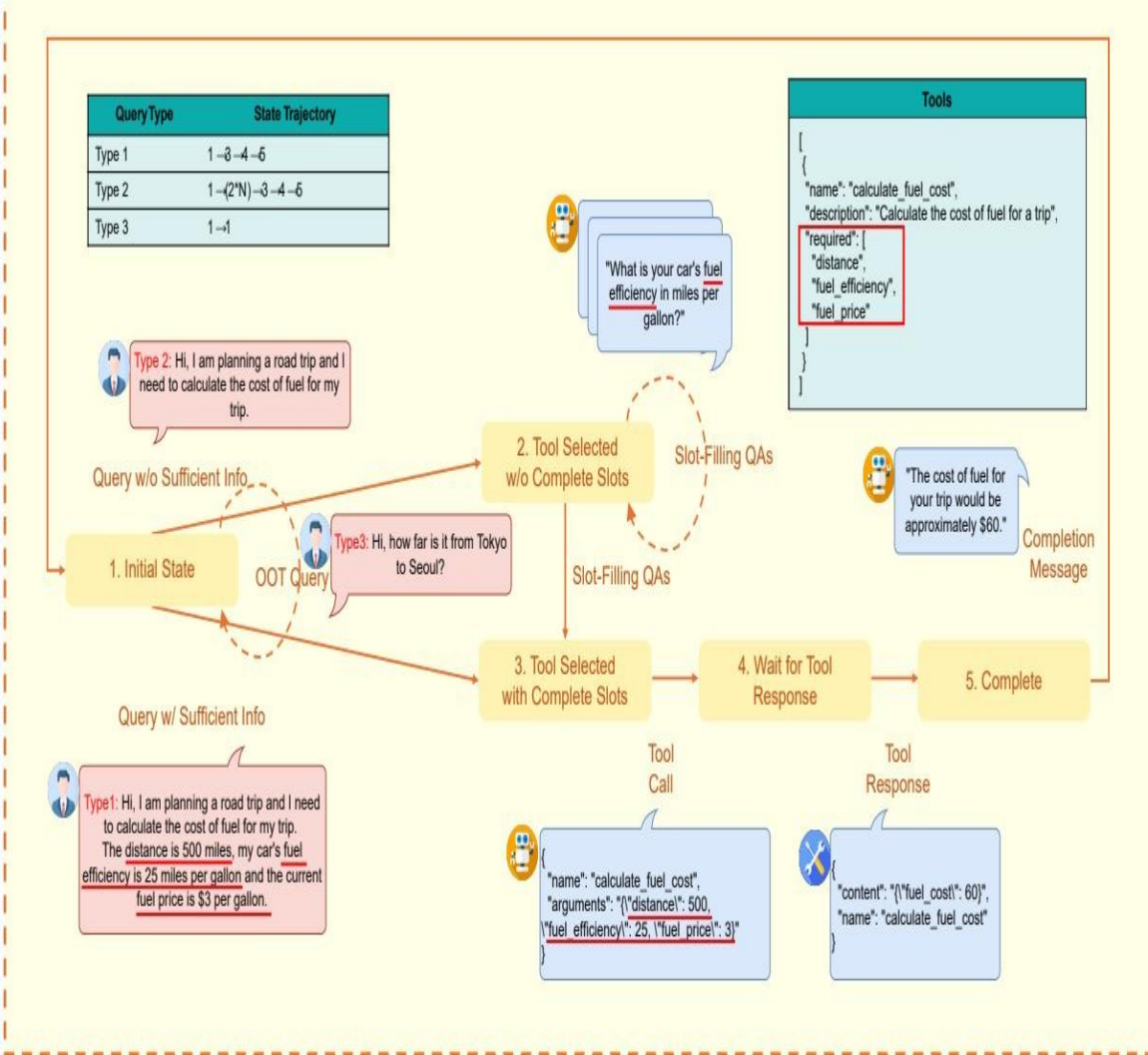
Failure to Reject Out-of-Scope Requests

When no suitable tool exists, SFT models frequently force a tool call instead of responding with a graceful rejection message.

DiaTool-DPO addresses all three failure modes by framing dialogue control as an MDP and automatically constructing chosen/rejected trajectory pairs for DPO training.

Modeling TA-LLM Dialogue as a 5-State Markov Decision Process

Each turn, the model transitions between unobserved internal states



S1 Initial State

No history. Out-of-scope requests return rejection and stay here.

S2 Tool Selected w/o Complete Slots

Tool identified but required parameters are missing — slot-filling QA occurs here.

S3 Tool Selected w/ Complete Slots

Both tool selection and all argument values are determined.

S4 Wait for Tool Response

Tool call executed; model awaits results from the external tool.

S5 Complete

Tool results received; completion message generated for the user.

Three Query Types Define Distinct State Trajectories

Every user interaction maps to one of three canonical paths through the MDP

Type	State Trajectory	Condition	Example
Type 1	$1 \rightarrow 3 \rightarrow 4 \rightarrow 5$	All required arguments present in the initial query	"Book a flight from NYC to LAX on Dec 12"
Type 2	$1 \rightarrow (2 \times N) \rightarrow 3 \rightarrow 4 \rightarrow 5$	Query lacks one or more required arguments; N slot-filling rounds needed	"Book a flight" $\rightarrow$ model asks: destination? date? passengers?
Type 3	$1 \rightarrow 1$	Requested functionality is unavailable — no matching tool exists	"Translate this text" when no translation tool is registered

State Flow Visualization

Type 1

$S1 \rightarrow S3 \rightarrow S4 \rightarrow S5$

Direct path — no slot-filling

Type 2

$S1 \rightarrow S2 (\times N) \rightarrow S3 \rightarrow S4 \rightarrow S5$

Iterative slot-filling loop

Type 3

$S1 \rightarrow S1$

Rejection loop — stays at Initial

Automatic Paired Trajectory Construction Without Human Labels

Chosen (correct) vs. Rejected (incorrect) trajectories generated from the MDP structure

Query	Chosen Traj.	Rejected Traj.	Learning Lesson	Count	Difficulty
Type 1	1→3→4→5	1→2→3→4→5	Prevent redundant slot-filling	2,089	Easy
Type 1	1→3→4→5	1→(2×N)→3→4→5	Prevent redundant slot-filling	562	Hard
Type 1	1→3→4→5	1→(2×M)→3→4→5	Prevent redundant slot-filling	2,530	Hard
Type 1	1→3→4→5	1→1	Tool call accept	2,090 / 562	Easy/Hard
Type 2	1→2→3→4→5	1→3→4→5	Prevent slot hallucination	2,089	Easy
Type 2	1→(2×N)→3→4→5	1→3→4→5	Prevent slot hallucination	562	Hard
Type 2	1→(2×N)→3→4→5	1→(2×M)→3→4→5	Prevent slot hallucination	2,530	Hard
Type 2	—	1→1	Tool call accept	2,089 / 562	Easy/Hard
Type 3	1→1	1→3→4	Tool call reject	567	Hard
Type 3	1→1	1→(2×N)→3→4	Tool call reject	562	Hard

SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

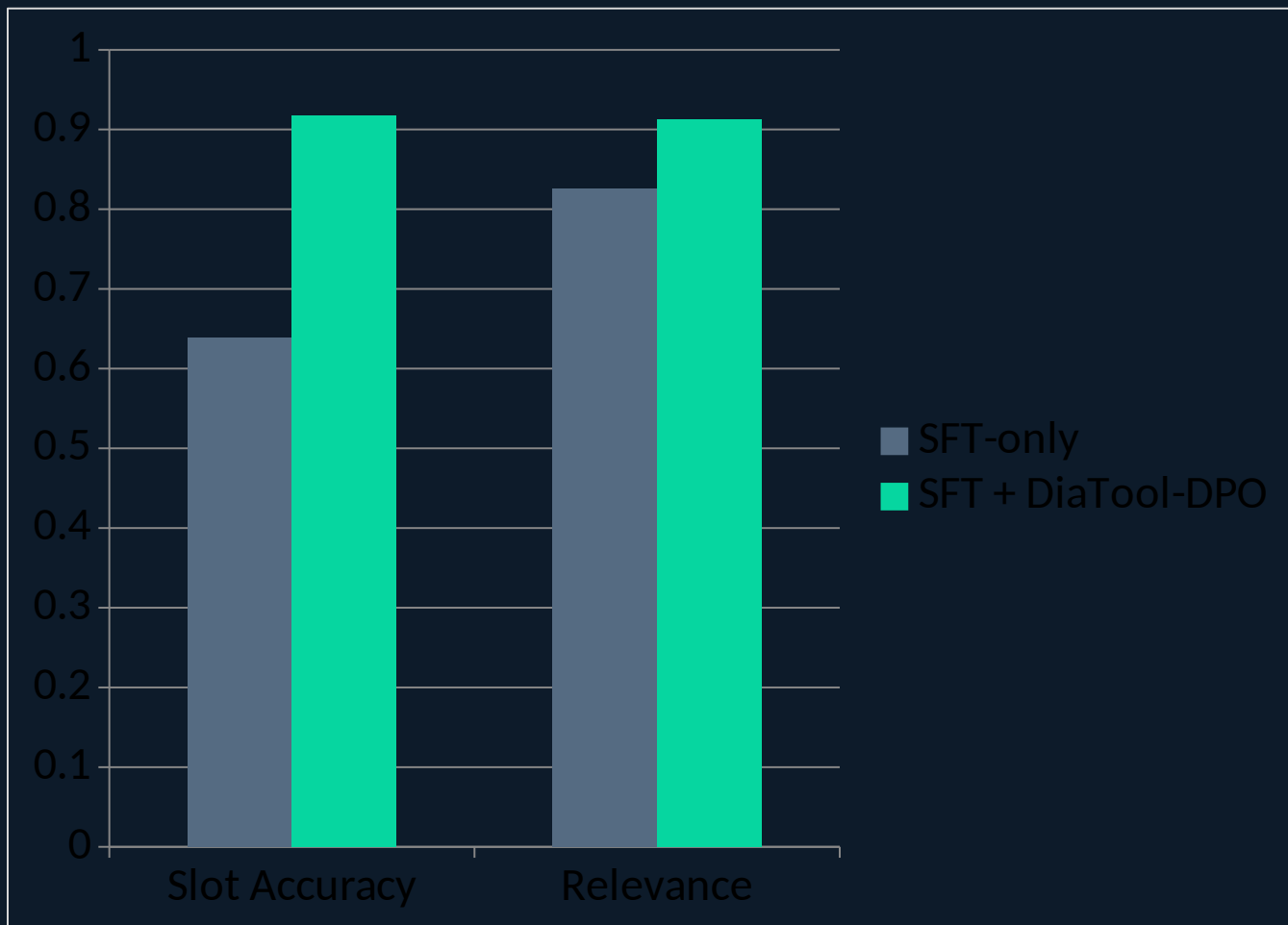
FunctionChat-Bench evaluation — bold = best per model group

Model	Method	Call	Completion	Slot	Relevance	Micro Avg	Macro Avg
Prop.-8B	DPO-only	0.314	0.700	0.833	0.609	0.575	0.614
Prop.-8B	SFT-only	0.900	0.916	0.694	0.913	0.870	0.856
Prop.-8B	SFT+DPO	0.886	0.929	0.833	0.826	0.884	0.868★
Prop.-3.1B	DPO-only	0.357	0.551	0.528	0.391	0.455	0.457
Prop.-3.1B	SFT-only	0.771	0.817	0.750	0.826	0.790	0.791
Prop.-3.1B	SFT+DPO	0.743	0.871	0.833	0.826	0.765	0.818★
LLaMA-3-8B	DPO-only	0.029	0.449	0.056	0.261	0.205	0.199
LLaMA-3-8B	SFT-only	0.843	0.957	0.639	0.826	0.844	0.816
LLaMA-3-8B	SFT+DPO	0.857	0.929	0.917	0.913	0.905	0.904★

★ Best Macro Avg per model group | DPO-only = DiaTool-DPO without SFT base

LLaMA-3-8B: +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

Adding DPO on top of SFT yields dramatic gains on the hardest dialogue-control tasks



Slot Accuracy

0.639 → 0.917 | +43.5%

Prevents slot hallucination — model now asks for missing parameters rather than fabricating values.

Relevance (Tool Rejection)

0.826 → 0.913 | +10.5%

Model correctly declines out-of-scope requests instead of forcing an inappropriate tool call.

Near GPT-4o Level

~94.8% info gathering | ~91% rejection

SFT + DiaTool-DPO on open 8B model approaches GPT-4o performance on key dialogue-control tasks.

Key Takeaways: RL-Based Dialogue Control for Production TA-LLMs

Summary of findings and implications for deploying tool-augmented language models

MDP Formulation Enables Systematic Preference Pairs

Defining 5 internal states and 3 query types creates a structured framework for automatically generating chosen/rejected trajectory pairs — no human annotation required.

DPO Complements SFT — Cannot Replace It

DiaTool-DPO-only (without SFT) performs poorly across all models. DPO refines dialogue control but requires a strong SFT base for basic tool-use capability.

Largest Gains on the Hardest Sub-Tasks

Slot-filling accuracy (+43.5%) and tool-call rejection (+10.5%) see the biggest improvements — precisely the tasks where SFT-only models hallucinate or fail to ask follow-ups.

Language-Agnostic and Broadly Applicable

Primary experiments in Korean generalize to English, confirming the method is architecture- and language-agnostic. Applicable to any production TA-LLM pipeline.

Path to GPT-4o-Level Performance on Smaller Open Models

SFT + DiaTool-DPO on LLaMA-3-8B reaches ~94.8% information gathering and ~91% tool rejection — approaching GPT-4o performance at a fraction of the inference cost.